

Project Report

Title: Diabetic Retinopathy Prediction Using Image Processing and Machine Learning

Author: Akshaya Chavan

Date: 10 February 2026

1. Introduction

Diabetic Retinopathy (DR) is a common complication of diabetes that affects the retina and can lead to vision loss if left untreated. Early detection is crucial to prevent severe outcomes. Manual screening of retinal images is time-consuming and requires expert ophthalmologists.

This project aims to develop a web-based application that allows users to upload retinal images and receive predictions about the presence and severity of diabetic retinopathy using a Convolutional Neural Network (CNN) model. In addition, the application displays related images from the same predicted category for user reference.

2. Objectives

To build a user-friendly web application for DR prediction. To integrate a CNN-based image classification model for retinal images. To provide prediction confidence levels. To display related images of the predicted class. To implement basic login and registration system for users.

3. Tools and Technologies Used

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python Flask Framework
Machine Learning	TensorFlow, Keras
Image Processing	Keras load_img, img_to_array, NumPy
File Handling	Python OS module, UUID Database (Future Use) SQLite/MySQL (currently placeholder)

4. Dataset Description

The dataset consists of retinal images categorized into five classes representing different stages of diabetic retinopathy:

Class	Description
0	No DR
1	Mild DR
2	Moderate DR
3	Severe DR
4	Proliferative DR

Training and testing split: Images are stored in Dataset/testing/<class_name>/ for testing purposes.

Each class contains multiple retinal images in PNG/JPG format.

5. System Architecture

The system has three main components:

5.1 Frontend

Home page: Introduction and navigation.

About page: Details about diabetic retinopathy and project objective.

Predict page: Allows image upload and displays results.

Login/Register pages: Allows user authentication (placeholder).

Navbar and Footer: Consistent across all pages.

5.2 Backend

Flask Application: Handles HTTP requests and routing.

File Upload Handling: Uploaded images are saved in uploads/ folder.

Prediction Processing: Images resized to 224x224 and normalized.

Model Integration: CNN model loaded from Vgg16-diabetes-best.h5 for classification.

5.3 Machine Learning Model

Model Type: Convolutional Neural Network (VGG16 architecture).

Input: Preprocessed 224x224 retinal images.

Output: Predicted class (0–4) and confidence score.

Related Images: Random selection of 4–5 images from the predicted class folder displayed to user.

6. Project Workflow

User uploads an image on the /predict page.

Flask backend checks file type and saves it securely in uploads/.

The image is preprocessed: Resized to 224x224 pixels Converted to NumPy array Normalized to range [0,1] Expanded dimensions for model input

Prediction:

Image passed to CNN model Predicted class determined using argmax Confidence calculated using softmax output

Related Images:

Folder corresponding to predicted class is scanned Random 4–5 images selected and displayed

Result Rendering:

Uploaded image, prediction, confidence, and related images displayed on predict.html

User Authentication:

Basic login and registration implemented

Placeholder authentication: username admin, password admin

Future integration with database for real authentication

7. Folder Structure

D:\Diabetic Project\

```
|
|
|─ app.py          # Flask application
|─ model\
|   └─ Vgg16-diabetes-best.h5 # Trained CNN model
|─ templates\      # HTML templates
|   └─ home.html
|   └─ about.html
|   └─ predict.html
|   └─ login.html
|   └─ register.html
|─ static\
|   └─ uploads\      # Uploaded images
|   └─ Dataset\testing\0\ # Class 0 images
|   └─ Dataset\testing\1\
|   └─ Dataset\testing\2\
|   └─ Dataset\testing\3\
|   └─ Dataset\testing\4\
|   └─ stylee.css    # CSS styling
|   └─ login1.png    # Login avatar
|   └─ regester1.png # Register avatar
|   └─ avatar.png
```

8. Code Highlights

8.1 Flask Routes

/ → Home

/about → About

/predict → Upload & Predict

/login → Login

/register → Register

/display/<filename> → Serve uploaded image

8.2 Image Prediction

```
img = load_img(file_path, target_size=(224,224))
```

```
img_array = img_to_array(img)/255.0
```

```
img_array = np.expand_dims(img_array, axis=0)
```

```
preds = model.predict(img_array)
```

```
pred_class = class_names[np.argmax(preds[0])]
```

```
confidence = np.max(preds[0])*100
```

8.3 Related Images

```
class_folder = os.path.join(DATASET_FOLDER, pred_class)
```

```
imgs = [img for img in os.listdir(class_folder) if allowed_file(img)]
```

```
random.shuffle(imgs)
```

```
related_imgs_paths = imgs[:5]
```

9. User Interface Design

Modern, clean, and responsive design using CSS

Navbar for navigation

Footer with copyright

Image preview before upload

Related images display below prediction

Login/Register cards with avatars

10. Future Enhancements

Integrate database (SQLite/MySQL) for real user authentication.

Expand dataset for more accurate predictions.

Add multi-class severity levels visualization.

Add mobile-friendly responsive design.

Implement email notifications for registered users with prediction results.

11. Conclusion

The Diabetic Retinopathy Prediction Web App demonstrates the integration of Machine Learning models with web technologies to provide a user-friendly solution for early DR detection. The project is highly scalable, and with future enhancements like real authentication and larger datasets, it can serve as a practical tool in healthcare for early DR screening.

12. References

Keras Documentation – <https://keras.io>

TensorFlow Documentation – <https://www.tensorflow.org>

Flask Documentation – <https://flask.palletsprojects.com>

Diabetic Retinopathy Dataset – Kaggle DR Dataset